

Khalid Mohamed

+20 109 755 2781 | kh.mohamed.dev@gmail.com | github.com/khalidua | linkedin.com/in/khalidmodev

EDUCATION

Zewail City of Science and Technology

Bachelor of Science (B.Sc.), Computer Science & Artificial Intelligence

Expected May 2027

GPA: 3.88/4.00

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Software Engineering, Database Systems, Operating Systems, Computer Networks

EXPERIENCE

Junior Teaching Assistant – Data Structures & Algorithms / OOP

Fall 2025 – Spring 2026

Zewail City of Science and Technology

- Served as Junior TA for Data Structures & Algorithms and Object-Oriented Programming in C++, supporting 60+ students
- Led weekly lab sessions covering recursion, trees, linked lists, sorting algorithms, complexity analysis, and object-oriented programming concepts
- Held weekly office hours supporting students in debugging C++ implementations of data structures and algorithms

PROJECTS

Social Networking Platform

[GitHub](#)

Go, PostgreSQL, Redis, Kafka, Docker, Kubernetes

- Architected a microservices-based social networking platform comprising 6 independently deployable services with API Gateway routing and event-driven communication
- Implemented OAuth2 authentication, JWT authorization, Redis caching, Kafka messaging, and database-per-service architecture using PostgreSQL
- Built containerized infrastructure using Docker and Kubernetes while integrating OpenSearch and Grafana for monitoring and observability

ZC League : Football League Management Platform

[Live Demo](#)

FastAPI, PostgreSQL, React, TypeScript, GitHub Actions, Azure

- Developed and deployed a full-stack football league management platform supporting tournament management, match scheduling, standings, notifications, and role-based workflows
- Designed RESTful APIs using FastAPI, implemented JWT authentication and RBAC, and modeled relational data using PostgreSQL
- Configured CI/CD pipelines with GitHub Actions and deployed production services to Azure Web Apps

ClipSphere : Short Video Social Platform

[GitHub](#)

Node.js, Express.js, MongoDB, Redis, Socket.IO, Docker

- Developed a full-stack short-video platform featuring real-time notifications, content management, and creator monetization workflows
- Implemented JWT authentication, RBAC authorization, Redis-backed processing, Socket.IO notifications, and Stripe payment integration
- Built a Dockerized multi-service architecture integrating MongoDB, MinIO object storage, Nginx, and scalable backend APIs

LEADERSHIP & ACHIEVEMENTS

- ECPC 2025 Participant : Solved algorithmic and data-structure problems under competitive programming contest constraints
- CIB Summer Program – The Green Leap (2025) : FinTech, AI, and banking innovation program

TECHNICAL SKILLS

Languages: Python, Go, C++, JavaScript, TypeScript, SQL, C#

Backend: FastAPI, Express.js, ASP.NET, Node.js, REST APIs, OpenAPI/Swagger

Frontend: React, Next.js, HTML, CSS, Tailwind CSS

Databases: PostgreSQL, SQL Server, MongoDB, Redis

Cloud & Infrastructure: Docker, Kubernetes, Azure, GitHub Actions, CI/CD

Software Engineering: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, Distributed Systems, Microservices

Tools: Git, GitHub, Postman, VS Code, Figma